



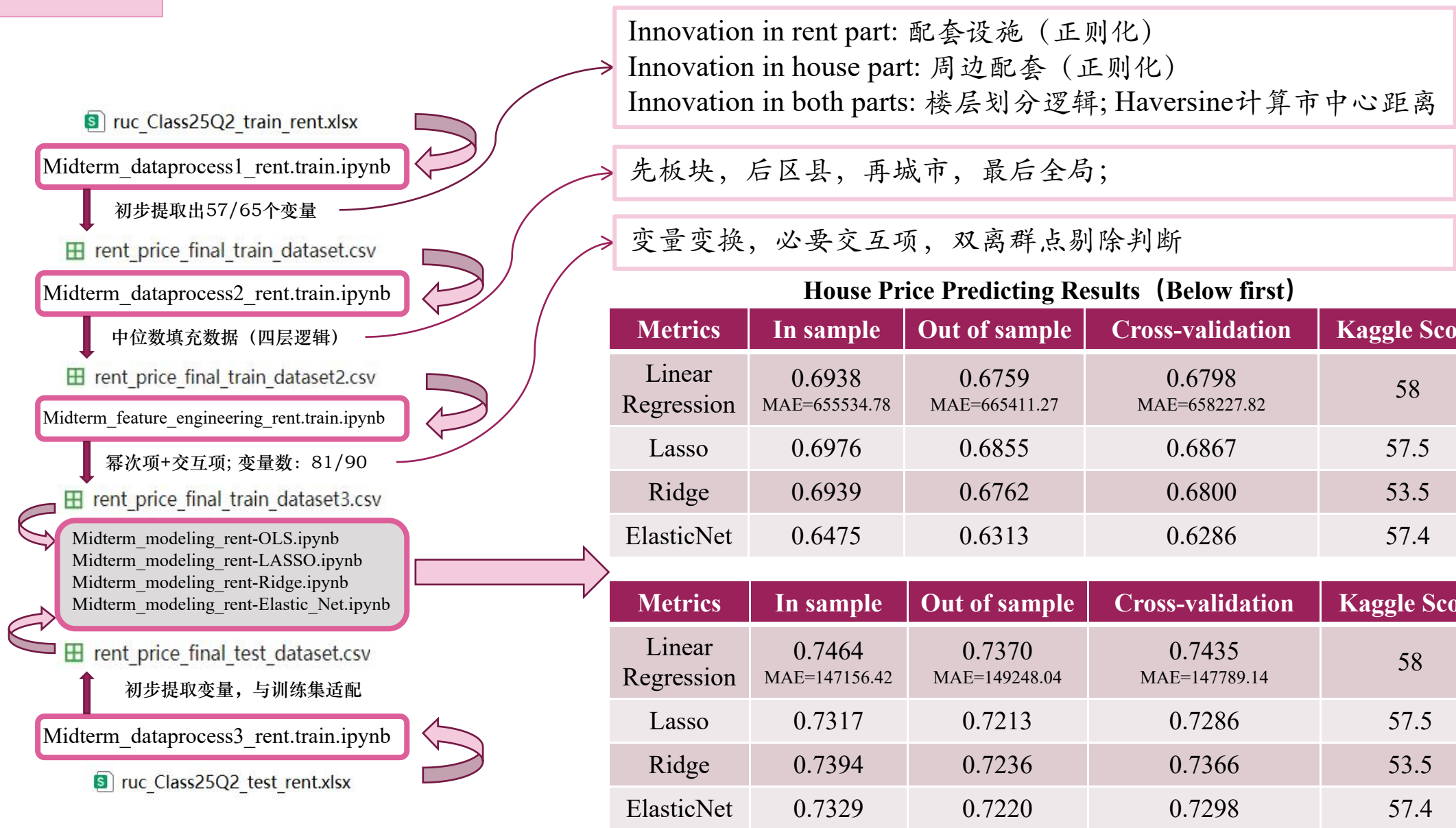
python midterm presentation

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From “Internet-Famous” to “Sustainably Popular”



Innovation in rent part: 配套设施（正则化）
Innovation in house part: 周边配套（正则化）
Innovation in both parts: 楼层划分逻辑; Haversine计算市中心距离

先板块，后区县，再城市，最后全局；

变量变换，必要交互项，双离群点剔除判断



■ Highlight the most important variable

Above all:Location and size

Plus some other things

Strategic Focus on Core Variables

Systematic Auxiliary Variable Processing

PRICE MODEL - MAE

Metrics	In sample	out of sample	Cross-validation
OLS	772,909.34	835,944.30	773,411.05
LASSO	835,278.84	918,747.90	835,347.44
Best Linear Model	772,883.05	835,901.06	773,378.71

RENT MODEL - MAE

Metrics	In sample	out of sample	Cross-validation
OLS	150,706.98	168,179.91	0.29
LASSO	204,079.16	236,058.66	0.38
Best Linear Model	150,706.98	168,179.91	0.29

Kaggle Score:

47.1



■ 创新点

3. 路畅宇

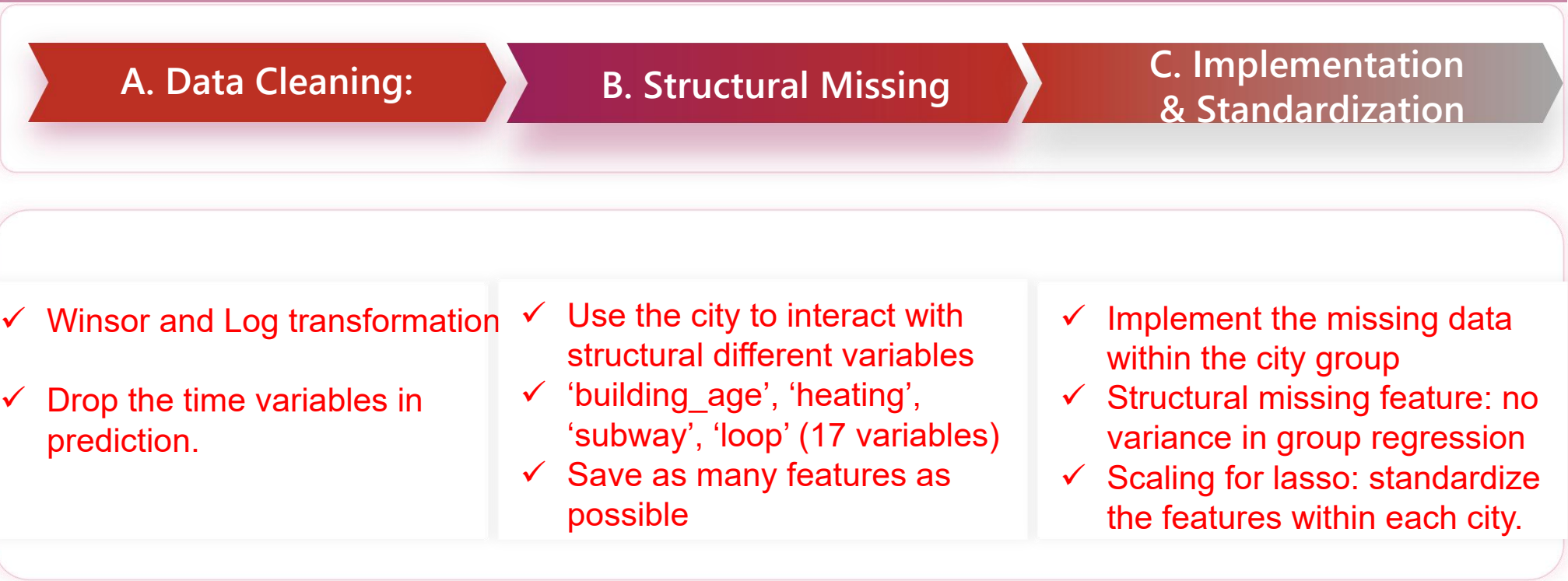
- 1) 利用单套房源各种特性做 K-Means，得到更细颗粒、质量—价格同质的“所属区域”，作为解释地段的变量
- 2) 用简洁的交互与时间哑变量，让线性模型捕捉“条件效应”和季节/档期规律。
- 3) 对于需要先挖参数的回归，选择先小样本寻优，后全量回归。

Metrics	In sample	out of sample	Cross-validation	Kaggle Score	Metrics	In sample	out of sample	Cross-validation	Kaggle Score
OLS	485454	484273	485610	66.4	OLS	108467	108784	108695	66.4
LASSO	485,891	484,906	485,843	65.9	LASSO	109,795	110,068	109,990	65.9
RIDGE	485,292	484,145	485,422	66.1	RIDGE	109,126	109,431	109,331	66.1
Elastic Net	496,829	496,353	496,849	60.2	Elastic Net	109,466	109,782	109,671	60.2
		(购房)					(租房)		
Metrics	In sample	out of sample	Cross-validation	Kaggle Score	Metrics	In sample	out of sample	Cross-validation	Kaggle Score
OLS	0.6473	0.6494	0.6472	66.4	OLS	0.7548	0.7578	0.7546	66.4
LASSO	0.6413	0.6422	0.6411	65.9	LASSO	0.7465	0.7492	0.7464	65.9
RIDGE	0.6472	0.6491	0.6471	66.1	RIDGE	0.751	0.7538	0.7508	66.1
Elastic Net	0.6283	0.6285	0.628	60.2	Elastic Net	0.7491	0.7518	0.749	60.2
		(买房)					(租房)		



■ Identify the variance across city (Interaction Method)

4. Jiamiao Xu
许家淼



PRICE MODEL - MAE				RENT MODEL - MAE				Kaggle Score:
Model	In Sample	Out of Sample	Cross-Validation	Model	In Sample	Out of Sample	Cross-Validation	
Lasso	687,509.15	672,358.92	688,658.84	OLS	161,839.70	160,167.64	162,035.88	
Ridge	687,461.22	672,238.30	688,618.69	Lasso	161,837.49	160,150.97	162,036.57	
ElasticNet	688,874.87	674,231.91	689,974.85	Ridge	161,839.70	160,167.64	162,035.88	34.4